A

Mini Project Report on

Chameleon

Submitted in partial fulfillment of the requirements for the degree

Second Year Engineering – Computer Science and Engineering (Data Science)

by

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CERTIFICATE

This is to certify that the Mini Project report on **CHAMELEON – A Stealth File Vault Android Application with AES-256-GCM Encryption & Identity Camouflage** has been submitted by **Abhishek Pal, Kshitij Nalawade, and Satyam Pandey** who are bonafide students of A. P. Shah Institute of Technology, Thane as a partial fulfillment of the requirement for the degree in Computer Science & Engineering, during the academic year 2025-2026 in the satisfactory manner as per the curriculum laid down by University of Mumbai.

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Chapter 1

Introduction

This chapter presents the background, problem statement, objectives, and scope of the Chameleon project — a stealth Android file vault application designed to protect sensitive personal media through advanced encryption and identity camouflage techniques.

1.1. Problem Statement

Smartphones store highly personal media files including photos, videos, and documents with no strong encryption protection at rest. Once a device is unlocked, virtually any installed application can potentially access the device gallery. Existing vault applications on the market are visually identifiable on the home screen and immediately raise suspicion when discovered by an adversary or unauthorized user. Furthermore, there is no credible deniability mechanism in conventional vault apps — an adversary who finds the app can compel the user to reveal the contents.

1.2. Objectives

- To develop a dual-PIN system with a real vault and a decoy vault to provide plausible deniability under coercion or observation.
- To allow users to import, view, export, photos and videos within a seamless in-app experience.
- To implement app icon switching (Calculator / Clock / Weather).
- To provide a decoy vault feature that enables plausible deniability by allowing users to display harmless content while keeping sensitive data securely hidden.

1.3. Scope

- Can be used by individuals to protect sensitive personal photos and videos from unauthorized viewing on their Android devices.
- Applicable to journalists, activists, lawyers, and professionals who require confidential encrypted media storage on mobile devices.
- Can be deployed on any Android 5.0+ device (API 21 and above), covering over 98% of all active Android smartphones.
- Applicable in personal security, domestic privacy, enterprise mobile data protection, and digital rights environments.
- The decoy vault is particularly valuable in regions with limited digital privacy rights, enabling plausible deniability.

Chapter 2
Proposed System

2.1. Features and Functionality

1: Calculator Disguise Interface

The app presents a fully functional calculator on the home screen. Entering the real PIN (e.g., '1234=') unlocks the real vault; entering the decoy PIN opens a harmless decoy vault. To any observer, it appears to be a normal calculator in use, with no visible indication of a hidden vault.

2: AES-256-GCM Encryption Engine

Every imported file is encrypted individually using AES-256-GCM with a unique random IV per file. Files are stored as '.enc' blobs in app-private storage. Decryption happens only in-memory during viewing — no plaintext file is ever written to disk, eliminating the risk of forensic extraction.

3: Dual-PIN Vault (Real & Decoy)

Two separate PINs unlock two distinct vaults. The real vault contains the user's genuine sensitive media. The decoy vault holds harmless files configured by the user. An adversary can be shown the decoy vault with no evidence of the real vault's existence, providing true plausible deniability.

4: Intruder Selfie Capture

After 3 consecutive failed PIN attempts, the front-facing camera silently captures a photo of the person attempting access. The photo is timestamped, geo-tagged, and stored as an encrypted file in the real vault — completely invisible to the intruder. This provides forensic evidence for identifying unauthorized access attempts.

5: Dynamic App Icon Switching

Users can switch the home screen icon between Calculator, Amazon Shopping, or the default icon from within the vault. Switching uses Android Activity Alias and PackageManager APIs — no reinstallation is needed and no data is lost during the icon change.

6: Encrypted Trash & Secure Deletion

Deleted files are moved to an encrypted trash bin before permanent removal. Permanent deletion performs a multi-pass cryptographic wipe (overwriting with random bytes) to prevent forensic data recovery from device storage.

2.2. Block Diagram

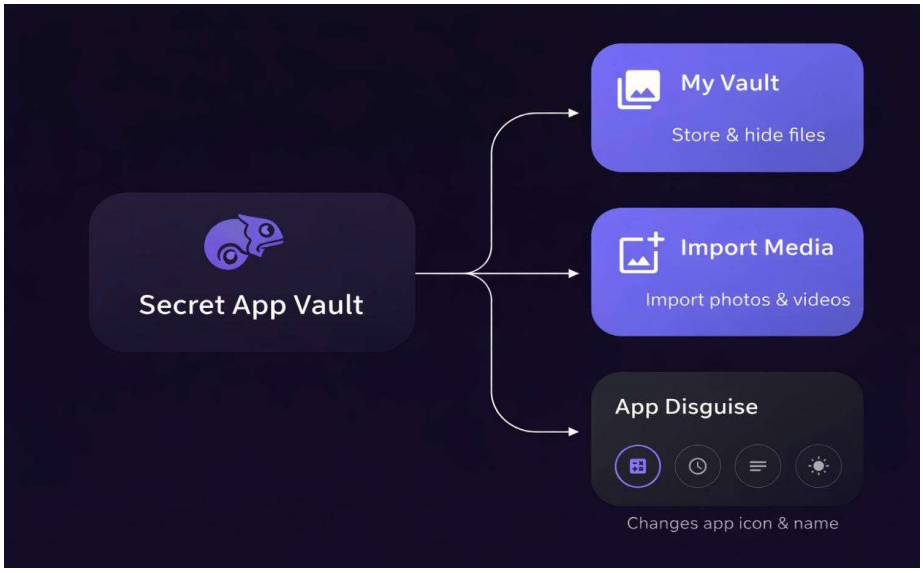

Figure 2.1 – System Block Diagram

Secret App Vault is a privacy-focused mobile application that allows users to securely hide their personal files and apps. The “My Vault” feature helps store important photos, videos, and documents safely. With the “Import Media” option, users can easily transfer data from their gallery into the vault. The app disguise feature lets users change the app’s icon and name to keep it hidden. It provides protection against unauthorized access. Overall, it is a useful tool for enhancing digital privacy and security.

Chapter 3

Project Outcomes

The successful implementation of the Chameleon application aims to achieve the following tangible outcomes:

- **Encrypted Media Import** – User can import photos and videos from the device gallery into the encrypted vault, where they are immediately and individually encrypted using AES-256-GCM.
- **In-Memory Decryption** – User can securely view encrypted images and videos inside the app; decryption is entirely in-memory, and no plaintext file is ever written to disk.
- **Plausible Deniability** – User can show the decoy vault with the decoy PIN; the real vault remains completely hidden and inaccessible under coercion.
- **Intruder Selfie Capture** – The app silently captures a front-camera photo of any intruder who fails the PIN 3 times and stores it encrypted in the real vault with timestamp and geo-tag.
- **Dynamic Icon Switching** – User can switch the home screen app icon between Calculator and Amazon Shopping without reinstalling the app or losing any vault data.
- **Encrypted Backup & Restore** – User can perform a full encrypted backup as a ZIP file and restore it on any Android device running Chameleon.
- **Secure Deletion** – User can permanently delete files using a multi-pass cryptographic wipe; forensic recovery of deleted files is not possible.

Chapter 4

Technology Stack & Software Requirements

4.1. Hardware Requirements

- **Processor:** Minimum Quad-core 2.0 GHz (for development machine)
- **RAM:** 8 GB Minimum (16 GB recommended for Android Studio / VS Code)
- **Device:** Android 5.0+ smartphone (API 21 and above) with front-facing camera
- **Storage:** Minimum 2 GB internal storage for app-private encrypted vault

4.2. Software / Technology Stack

Frontend / UI Framework: Flutter 3.10+ (Dart) — Cross-platform mobile UI toolkit; single codebase for Android. Flutter ScreenUtil is used for responsive layouts across different screen sizes and resolutions.

Encryption & Security: AES-256-GCM via Dart cryptography package. Flutter Secure Storage for PIN and key storage. Local Auth package for biometric (fingerprint) authentication support.

Media & Camera: Photo Manager (gallery import/export), Camera plugin (intruder selfie capture), Video Player plugin (in-app playback from temp file — fully offline).

File System & Backup: path_provider (app-private storage), archive package (ZIP backup/restore), file_picker (backup import), MIME type detection via mime package.

Native Android Layer: Kotlin (MainActivity + MethodChannel) for PackageManager-based icon switching. Minimum SDK: Android 5.0 (API 21). Target SDK: Android 14 (API 34).

4.3. Operating System

- **Development OS:** Windows 10/11, macOS, or Linux
- **Target Platform:** Android 5.0+ (API 21+)
- **IDE:** Android Studio or Visual Studio Code with Flutter/Dart plugins

Chapter 5
Project Design

5.1. System Architecture

Chameleon follows a layered architecture to ensure security isolation, scalability, and ease of maintenance. The system is divided into three primary layers:

- **Presentation Layer (Flutter):** Handles the UI components, calculator disguise interface, PIN entry, and in-app media viewer.
- **Logic Layer (Dart BLoC/Provider):** Manages state, encryption/decryption operations, dual-PIN routing, intruder detection logic, and icon switching.
- **Data Layer (File System & Secure Storage):** Manages encrypted .enc file blobs in app-private storage, ZIP backup archives, and secure PIN/key storage.

5.2. Work Flow

Figure 5.1 – Work Flow Diagram

This diagram represents the workflow of the Chameleon – Stealth File Vault application. It starts when the user launches the app, which appears as a normal calculator interface. After entering a PIN, the system decides whether to open the real vault, a decoy vault, or trigger an intruder alert. The real vault provides secure access to encrypted files, while the decoy vault shows fake data for protection. If an incorrect PIN is entered, the app detects the intruder and captures security details. Overall, the system ensures strong data privacy through encryption and smart disguise techniques

Chapter 6

Project Scheduling

As per the semester requirements (January to April 2026), the project was carried out using a well-structured and systematic schedule divided into two phases — Conception & Initiation and Design & Implementation — with tasks assigned to Kshitij Nalawade, Abhishek Pal, and Satyam Pandey. The Gantt Chart below provides a clear overview of task distribution and timely completion within the academic timeline.

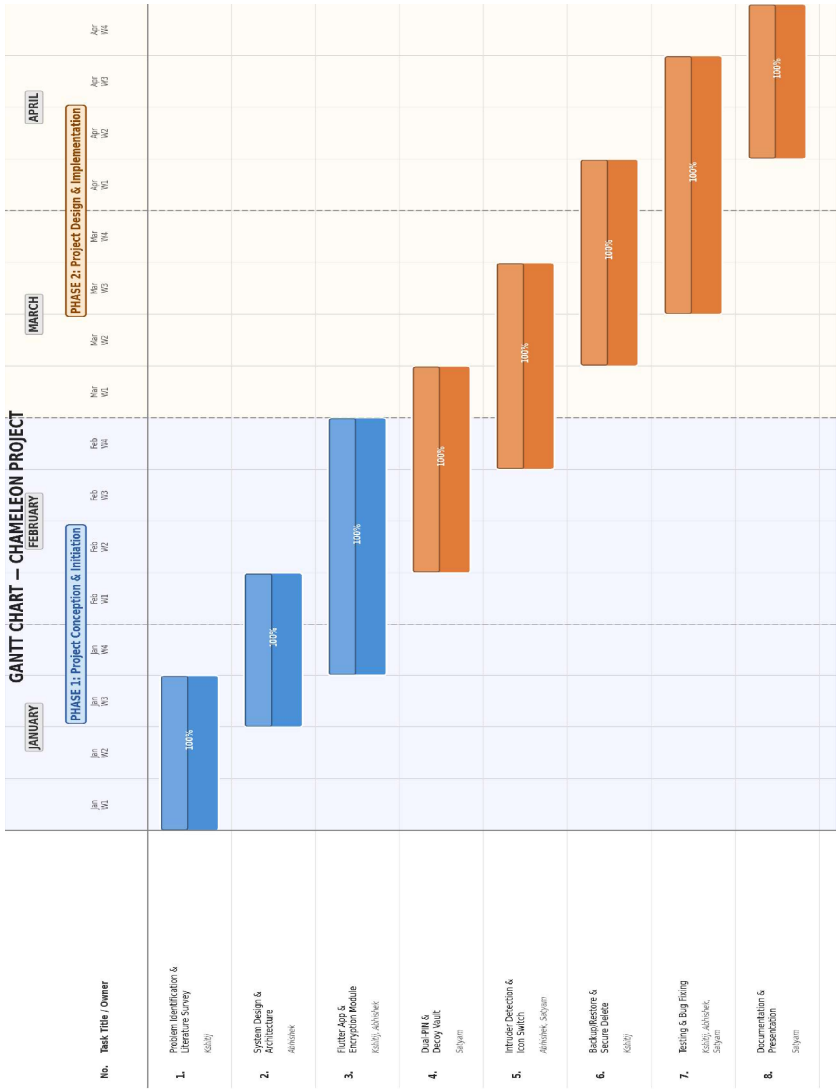

Figure 6.1 – Gantt Chart Diagram

Chapter 7

Results

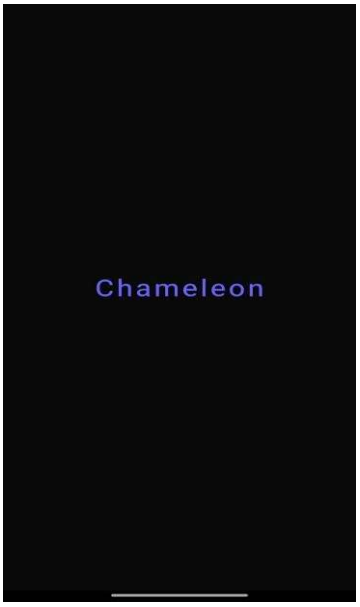

Figure. 7.1 – Chameleon Splash / Launch Screen

The Chameleon splash screen is intentionally minimalistic — it displays only the application name 'Chameleon' in a subtle blue typeface against a completely black background. This design choice is deliberate: there is no logo, no vault icon, no indication of encryption, and no hint of a hidden security feature.

To any casual observer or an adversary who picks up the device, the application appears to be a plain, unremarkable utility. This is the first layer of identity camouflage that Chameleon employs — the launch screen itself reveals nothing about the application's true purpose.

The splash screen is displayed briefly (approximately 1–2 seconds) before automatically transitioning to the PIN entry screen. No internet connection is required at this stage, and no data is transmitted externally. The screen also serves as an initialization point where the application loads encryption keys securely from Flutter Secure Storage into memory.

This output confirms that the app icon disguise feature is functioning correctly — the home screen icon (Calculator / Amazon Shopping / Clock) successfully launches the Chameleon vault without exposing its identity through the splash screen.

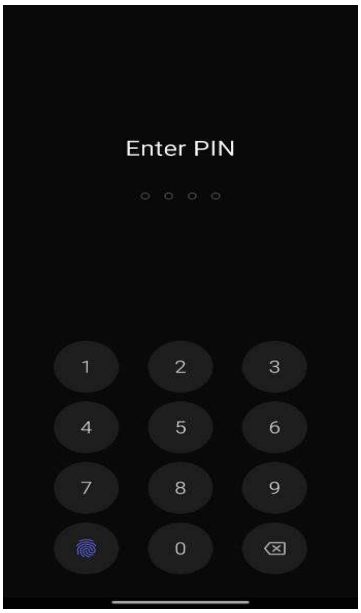

Fig. 7.2 – Dual-PIN Authentication Screen

The PIN entry screen is the primary authentication gateway to the Chameleon vault. It presents a clean, dark-themed numeric keypad with four dot indicators at the top that fill in as the user enters each digit. A fingerprint / biometric icon is also visible on the bottom-left of the keypad, supporting biometric authentication via the Flutter Local Auth package.

Chameleon implements a dual-PIN system that is central to its plausible deniability architecture. When the user enters the Real PIN, the application decrypts and displays the actual vault containing all sensitive encrypted media. When the user enters the Decoy PIN, a completely separate, harmless vault is displayed — one that contains only benign files pre-configured by the user.

An adversary who compels the user to unlock the app under coercion will see only the decoy vault. There is no visible UI difference between the real and decoy vault interfaces — both look identical to the user. The real vault's existence cannot be detected from the decoy session, as no metadata, file counts, or storage references are shared between the two vault spaces.

Additionally, after 3 consecutive incorrect PIN attempts, the application silently activates the front-facing camera, captures a photograph of the person attempting access, geo-tags and timestamps it, and stores it as an AES-256-GCM encrypted file in the real vault — completely invisible to the intruder.

Fig. 7.3 – Vault Dashboard & App Disguise Panel

After successful authentication, the user is presented with the main Vault Dashboard. The top bar displays the app name 'Chameleon' along with a settings icon and a logout button. The dashboard is divided into two primary functional sections: My Vault and Import Media.

My Vault is the core feature — tapping it opens the encrypted media gallery where all imported photos and videos are displayed as thumbnails. Each file is stored as an AES-256-GCM encrypted '.enc' blob in app-private storage. Decryption occurs entirely in-memory during viewing; no plaintext file is ever written to disk, ensuring that forensic extraction tools cannot recover unencrypted content.

Import Media allows the user to select photos and videos from the device gallery and import them into the vault. Upon import, each file is individually encrypted with a unique random Initialization Vector (IV) using AES-256-GCM. The original file can optionally be deleted from the gallery after import.

The App Disguise panel at the bottom is one of the most distinctive features of Chameleon. It displays four icon options — Calculator, Clock, Notes, and Weather. Selecting any of these dynamically changes the home screen icon and app name using Android's Activity Alias and PackageManager APIs. No reinstallation is required, and no vault data is affected by the icon switch. This ensures that the application remains undetectable on the home screen at all times.

Chapter 8

Conclusion

Chameleon addresses a critical gap in mobile privacy by providing a fully stealth, encryption-grade file vault for Android users. Unlike conventional vault applications that are immediately identifiable and raise suspicion, Chameleon operates under the disguise of a fully functional Calculator or Amazon Shopping application — providing zero visual indication of a hidden vault on the home screen.

The application's AES-256-GCM encryption engine ensures that all media files stored within the vault are cryptographically secured at rest, with decryption occurring exclusively in-memory during viewing. The dual-PIN architecture provides a robust plausible deniability mechanism, allowing users to reveal a harmless decoy vault under coercion while keeping the real vault completely hidden and inaccessible.

Additional features such as intruder selfie capture, dynamic icon switching via Android PackageManager APIs, multi-pass secure deletion, and encrypted ZIP backup/restore collectively make Chameleon a comprehensive privacy solution for individuals who require strong, credible protection for sensitive media on their mobile devices.

Built entirely on Flutter with a native Kotlin layer for Android-specific operations, Chameleon is compatible with Android 5.0+ devices (over 98% of active Android smartphones) and operates completely offline, requiring no internet access for any of its core features. Future work will include cloud-encrypted offsite backup, cross-device sync, and a secure document scanning module.

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